

AI Exploratory Test Report

Target Application: <https://www.calculator.net>
Report Date: March 23, 2026 at 13:46
Tool Version: AI Exploratory Test Engine v2.0
Pages Analyzed: 10
Total Test Cases Generated: 56
Valid Test Cases: 56
Rejected (Hallucinated): 0
Tests Executed: 56
Passed: 56
Failed (Bugs Found): 0

2. Execution Results Summary

56 PASSED

Tests behaved correctly

0 FAILED

Bugs found in application

3. Application Coverage Summary

Metric	Value
Pages Crawled	10
Pages with Interactive Elements	10
Input Fields Identified	75
Buttons Identified	55
Navigation Links	193
Dropdowns/Selects	0
Test Cases Generated	56
Validation Pass Rate	100.0%

4. Batch Generation Strategy

Tests were generated in 6 adversarial batches. Each batch focused on a distinct exploration category to guarantee structural diversity and maximize bug-finding potential.

Batch	Category	Focus
1	Input Torture and Field Poisoning	Edge-case values: long strings, special chars, min-1/max+1, empty required fields.
2	State Transitions and Form Abandonment	Partial fills, navigate-away-return, recalculate after change, state persistence.
3	Boundary Value Analysis	Exact min, max, zero, +1/-1 boundaries across every numeric input field.
4	Multi-Step Navigation Chaos	Fill on page A, jump to B, return to A -- does state survive?
5	Mode Switching and UI State Dances	Toggle modes mid-fill, switch tabs, check value retention after mode change.
6	Error Recovery and Validation Resilience	Submit empty, fix one field, resubmit; clear after success; recovery flow testing.

5. Test Coverage Classification

Category	Count
General Exploratory Tests	21
Boundary Tests	15
Negative Input Tests	7
Validation Enforcement Tests	5
Zero/Empty Input Tests	3
State Persistence Tests	3
Navigation & Cross-Page Tests	2

Test cases are automatically classified based on their goals and step patterns. Categories are not mutually exclusive; each test is assigned to its primary category.

6. Validated Test Cases

56 test cases passed anti-hallucination validation. All referenced UI elements exist in the application snapshot.

Each test is marked PASSED (green) or FAILED (red) based on execution results.

VT-01 | Test input field with extremely large value [PASSED]

Page: /loan-calculator.html

Target: c2loanamount

Type: Boundary Tests

1. Navigate to '/loan-calculator.html'
2. Enter '999999999' into 'c2loanamount'
3. Click 'Calculate'

Expected: Validation failure

VT-02 | Test input field with special characters [PASSED]

Page: /mortgage-calculator.html

Target: chouseprice

Type: Negative Input Tests

1. Navigate to '/mortgage-calculator.html'
2. Enter '!@#\$\$' into 'chouseprice'
3. Click 'Calculate'

*Expected: Error message***VT-03 | Test input field with negative value [PASSED]**

Page: /auto-loan-calculator.html

Target: csaleprice

Type: Negative Input Tests

1. Navigate to '/auto-loan-calculator.html'
2. Enter '-1' into 'csaleprice'
3. Click 'Calculate'

*Expected: Validation failure***VT-04 | Test input field with whitespace-only value [PASSED]**

Page: /interest-calculator.html

Target: cstartingprinciple

Type: General Exploratory Tests

1. Navigate to '/interest-calculator.html'
2. Enter ' ' into 'cstartingprinciple'
3. Click 'Calculate'

*Expected: Error message***VT-05 | Test input field with extremely small value [PASSED]**

Page: /payment-calculator.html

Target: cloanamount

Type: Boundary Tests

1. Navigate to '/payment-calculator.html'
2. Enter '0.000001' into 'cloanamount'
3. Click 'Calculate'

*Expected: Validation failure***VT-06 | Test input field with cross-field contamination [PASSED]**

Page: /retirement-calculator.html

Target: cagenow

Type: General Exploratory Tests

1. Navigate to '/retirement-calculator.html'
2. Enter '123' into 'cagenow'
3. Copy value from 'cagenow' and paste into 'cretireage' with extra characters 'abc'
4. Click 'Calculate'

*Expected: Error message***VT-07 | Test input field with empty value [PASSED]**

Page: /amortization-calculator.html

Target: cloanamount

Type: Zero/Empty Input Tests

1. Navigate to '/amortization-calculator.html'
2. Enter '' into 'cloanamount'
3. Click 'Calculate'

Expected: Validation failure

VT-08 | Test input field with value exceeding max limit [PASSED]

Page: /loan-calculator.html

Target: c2loanamount

Type: Boundary Tests

1. Navigate to '/loan-calculator.html'
2. Enter '1000001' into 'c2loanamount'
3. Click 'Calculate'

*Expected: Error message***VT-09 | Test input field with alphanumerics [PASSED]**

Page: /mortgage-calculator.html

Target: cdownpayment

Type: Negative Input Tests

1. Navigate to '/mortgage-calculator.html'
2. Enter 'abc123' into 'cdownpayment'
3. Click 'Calculate'

*Expected: Validation failure***VT-10 | Test form submission with empty fields [PASSED]**

Page: /mortgage-calculator.html

Target: N/A

Type: Zero/Empty Input Tests

1. Navigate to '/mortgage-calculator.html'
2. Click 'Calculate'

*Expected: Validation error for missing fields***VT-11 | Test form submission with partial data [PASSED]**

Page: /mortgage-calculator.html

Target: chouseprice

Type: General Exploratory Tests

1. Navigate to '/mortgage-calculator.html'
2. Enter '100000' into 'chouseprice'
3. Click 'Calculate'

*Expected: Validation error for missing fields***VT-12 | Test form submission with valid data and then modify a field [PASSED]**

Page: /mortgage-calculator.html

Target: chouseprice

Type: General Exploratory Tests

1. Navigate to '/mortgage-calculator.html'
2. Enter '100000' into 'chouseprice'
3. Enter '20' into 'cdownpayment'
4. Enter '30' into 'cloanterm'
5. Enter '4' into 'cinterestrates'
6. Click 'Calculate'
7. Enter '5' into 'cinterestrates'
8. Click 'Calculate'

*Expected: Validation error for updated fields***VT-13 | Test form submission with one field empty at a time [PASSED]**

Page: /mortgage-calculator.html

Target: chouseprice

Type: Zero/Empty Input Tests

1. Navigate to '/mortgage-calculator.html'
2. Enter '100000' into 'chouseprice'
3. Enter '20' into 'cdownpayment'
4. Enter '30' into 'cloanterm'
5. Click 'Calculate'

Expected: Validation error for missing field

VT-14 | Test form submission with valid data and then submit again w [PASSED]

Page: /mortgage-calculator.html

Target: chouseprice

Type: General Exploratory Tests

1. Navigate to '/mortgage-calculator.html'
2. Enter '100000' into 'chouseprice'
3. Enter '20' into 'cdownpayment'
4. Enter '30' into 'cloanterm'
5. Enter '4' into 'cinterestrates'
6. Click 'Calculate'
7. Click 'Calculate'

*Expected: Validation error for duplicate submission***VT-15 | Test form submission with partial data and then submit again [PASSED]**

Page: /mortgage-calculator.html

Target: chouseprice

Type: State Persistence Tests

1. Navigate to '/mortgage-calculator.html'
2. Enter '100000' into 'chouseprice'
3. Click 'Calculate'
4. Enter '20' into 'cdownpayment'
5. Enter '30' into 'cloanterm'
6. Enter '4' into 'cinterestrates'
7. Click 'Calculate'

*Expected: Validation error for updated fields***VT-16 | Test form submission with valid data and then clear the form [PASSED]**

Page: /mortgage-calculator.html

Target: chouseprice

Type: General Exploratory Tests

1. Navigate to '/mortgage-calculator.html'
2. Enter '100000' into 'chouseprice'
3. Enter '20' into 'cdownpayment'
4. Enter '30' into 'cloanterm'
5. Enter '4' into 'cinterestrates'
6. Click 'Calculate'
7. Click 'Clear'
8. Click 'Calculate'

*Expected: Validation error for missing fields***VT-17 | Test form submission with valid data and then modify a field [PASSED]**

Page: /mortgage-calculator.html

Target: chouseprice

Type: General Exploratory Tests

1. Navigate to '/mortgage-calculator.html'
2. Enter '100000' into 'chouseprice'
3. Enter '20' into 'cdownpayment'
4. Enter '30' into 'cloanterm'
5. Enter '4' into 'cinterestrates'
6. Click 'Calculate'
7. Enter '5' into 'cinterestrates'
8. Click 'Clear'
9. Click 'Calculate'

Expected: Validation error for missing fields

VT-18 | Break loan calculator with minimum loan amount [PASSED]

Page: /loan-calculator.html

Target: cloanamount

Type: Boundary Tests

1. Navigate to '/loan-calculator.html'
2. Enter '0' into 'cloanamount'
3. Click 'Calculate'

*Expected: Invalid loan amount***VT-19 | Break mortgage calculator with maximum interest rate [PASSED]**

Page: /mortgage-calculator.html

Target: cinterestrate

Type: General Exploratory Tests

1. Navigate to '/mortgage-calculator.html'
2. Enter '100' into 'cinterestrate'
3. Click 'Calculate'

*Expected: Invalid interest rate***VT-20 | Break auto loan calculator with very large sale price [PASSED]**

Page: /auto-loan-calculator.html

Target: csaleprice

Type: Boundary Tests

1. Navigate to '/auto-loan-calculator.html'
2. Enter '9999999999' into 'csaleprice'
3. Click 'Calculate'

*Expected: Calculation error***VT-21 | Break interest calculator with very small monthly addition [PASSED]**

Page: /interest-calculator.html

Target: cmonthlyaddition

Type: Boundary Tests

1. Navigate to '/interest-calculator.html'
2. Enter '0.0001' into 'cmonthlyaddition'
3. Click 'Calculate'

*Expected: Rounding error***VT-22 | Break payment calculator with one below minimum loan term [PASSED]**

Page: /payment-calculator.html

Target: cloanterm

Type: Boundary Tests

1. Navigate to '/payment-calculator.html'
2. Enter '0' into 'cloanterm'
3. Click 'Calculate'

*Expected: Invalid loan term***VT-23 | Break retirement calculator with very large annual save [PASSED]**

Page: /retirement-calculator.html

Target: cannualsave

Type: Boundary Tests

1. Navigate to '/retirement-calculator.html'
2. Enter '9999999999' into 'cannualsave'
3. Click 'Calculate'

Expected: Calculation error

VT-24 | Break amortization calculator with zero interest rate [PASSED]

Page: /amortization-calculator.html

Target: cinterestrate

Type: Boundary Tests

1. Navigate to '/amortization-calculator.html'
2. Enter '0' into 'cinterestrate'
3. Click 'Calculate'

*Expected: Invalid interest rate***VT-25 | Break loan calculator with one above maximum loan term [PASSED]**

Page: /loan-calculator.html

Target: cloantermmonth

Type: General Exploratory Tests

1. Navigate to '/loan-calculator.html'
2. Enter '61' into 'cloantermmonth'
3. Click 'Calculate'

*Expected: Invalid loan term***VT-26 | Break mortgage calculator with very small down payment [PASSED]**

Page: /mortgage-calculator.html

Target: cdownpayment

Type: Boundary Tests

1. Navigate to '/mortgage-calculator.html'
2. Enter '0.0001' into 'cdownpayment'
3. Click 'Calculate'

*Expected: Rounding error***VT-27 | Break auto loan calculator with maximum incentive [PASSED]**

Page: /auto-loan-calculator.html

Target: cincentive

Type: General Exploratory Tests

1. Navigate to '/auto-loan-calculator.html'
2. Enter '1000000' into 'cincentive'
3. Click 'Calculate'

*Expected: Calculation error***VT-28 | Test rapid input submission on loan calculator [PASSED]**

Page: /loan-calculator.html

Target: cloanamount

Type: General Exploratory Tests

1. Navigate to '/loan-calculator.html'
2. Enter '10000' into 'cloanamount'
3. Enter '5' into 'cloanterm'
4. Click 'Calculate'
5. Enter '20000' into 'cloanamount'
6. Enter '10' into 'cloanterm'
7. Click 'Calculate'

*Expected: Validation failure for duplicate submissions***VT-29 | Test multi-step form interaction on mortgage calculator [PASSED]**

Page: /mortgage-calculator.html

Target: chouseprice

Type: General Exploratory Tests

1. Navigate to '/mortgage-calculator.html'
2. Enter '200000' into 'chouseprice'
3. Enter '20' into 'cdownpayment'
4. Click 'Calculate'
5. Click 'Clear'
6. Enter '300000' into 'chouseprice'

7. Enter '30' into 'cdownpayment'
8. Click 'Calculate'

Expected: Error message for invalid input

VT-30 | Test rapid submission on auto loan calculator [PASSED]

Page: /auto-loan-calculator.html

Target: csaleprice

Type: General Exploratory Tests

1. Navigate to '/auto-loan-calculator.html'
2. Enter '10000' into 'csaleprice'
3. Enter '5' into 'cloanterm'
4. Click 'Calculate'
5. Click 'Calculate'
6. Click 'Calculate'

Expected: Validation failure for duplicate submissions

VT-31 | Test required field submission on payment calculator [PASSED]

Page: /payment-calculator.html

Target: cloanamount

Type: Validation Enforcement Tests

1. Navigate to '/payment-calculator.html'
2. Enter '10000' into 'cloanamount'
3. Enter '5' into 'cloanterm'
4. Click 'Calculate'

Expected: No UI errors or crashes

VT-32 | Test clear and resubmit on interest calculator [PASSED]

Page: /interest-calculator.html

Target: cstartingprinciple

Type: General Exploratory Tests

1. Navigate to '/interest-calculator.html'
2. Enter '10000' into 'cstartingprinciple'
3. Enter '5' into 'cannualaddition'
4. Click 'Calculate'
5. Click 'Clear'
6. Enter '20000' into 'cstartingprinciple'
7. Enter '10' into 'cannualaddition'
8. Click 'Calculate'

Expected: Error message for invalid input

VT-33 | Test rapid input submission on retirement calculator [PASSED]

Page: /retirement-calculator.html

Target: cagenow

Type: General Exploratory Tests

1. Navigate to '/retirement-calculator.html'
2. Enter '30' into 'cagenow'
3. Enter '60' into 'cretireage'
4. Click 'Calculate'
5. Enter '40' into 'cagenow'
6. Enter '70' into 'cretireage'
7. Click 'Calculate'

Expected: Validation failure for duplicate submissions

VT-34 | Test multi-step form interaction on amortization calculator [PASSED]

Page: /amortization-calculator.html

Target: cloanamount

Type: General Exploratory Tests

1. Navigate to '/amortization-calculator.html'
2. Enter '10000' into 'cloanamount'
3. Enter '5' into 'cloanterm'
4. Click 'Calculate'
5. Click 'Clear'
6. Enter '20000' into 'cloanamount'
7. Enter '10' into 'cloanterm'
8. Click 'Calculate'

*Expected: Error message for invalid input***VT-35 | Test required field submission on loan calculator [PASSED]**

Page: /loan-calculator.html

Target: cloanamount

Type: Validation Enforcement Tests

1. Navigate to '/loan-calculator.html'
2. Enter '10000' into 'cloanamount'
3. Enter '5' into 'cloanterm'
4. Click 'Calculate'

*Expected: No UI errors or crashes***VT-36 | Test rapid submission on payment calculator [PASSED]**

Page: /payment-calculator.html

Target: cloanamount

Type: General Exploratory Tests

1. Navigate to '/payment-calculator.html'
2. Enter '10000' into 'cloanamount'
3. Enter '5' into 'cloanterm'
4. Click 'Calculate'
5. Click 'Calculate'
6. Click 'Calculate'

*Expected: Validation failure for duplicate submissions***VT-37 | Break loan calculator by rapid mode switching [PASSED]**

Page: /loan-calculator.html

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/loan-calculator.html'
2. Click 'Calculate'
3. Click 'Clear'
4. Click 'Calculate'
5. Click 'Search'

*Expected: No UI errors***VT-38 | Test interest calculator with invalid input [PASSED]**

Page: /interest-calculator.html

Target: cstartingprinciple

Type: Negative Input Tests

1. Navigate to '/interest-calculator.html'
2. Enter 'abc' into 'cstartingprinciple'
3. Click 'Calculate'

Expected: Validation failure

VT-39 | Rapidly switch between calculation modes on the loan calcula [PASSED]

Page: /loan-calculator.html

Target: N/A

Type: Navigation & Cross-Page Tests

1. Navigate to '/loan-calculator.html'
2. Click 'Calculate'
3. Click 'Clear'
4. Click 'Calculate'
5. Click 'Search'
6. Click 'Calculate'

*Expected: No UI errors***VT-40 | Break payment calculator by entering invalid loan term [PASSED]**

Page: /payment-calculator.html

Target: cloanterm

Type: Negative Input Tests

1. Navigate to '/payment-calculator.html'
2. Enter 'abc' into 'cloanterm'
3. Click 'Calculate'

*Expected: Validation failure***VT-41 | Test retirement calculator with extreme input values [PASSED]**

Page: /retirement-calculator.html

Target: cagenow

Type: Boundary Tests

1. Navigate to '/retirement-calculator.html'
2. Enter '1000' into 'cagenow'
3. Click 'Calculate'

*Expected: Error message***VT-42 | Rapidly apply and unapply a mode on the amortization calcula [PASSED]**

Page: /amortization-calculator.html

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/amortization-calculator.html'
2. Click 'Calculate'
3. Click 'Clear'
4. Click 'Calculate'
5. Click 'Search'
6. Click 'Clear'

*Expected: No UI errors***VT-43 | Test auto loan calculator with missing input values [PASSED]**

Page: /auto-loan-calculator.html

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/auto-loan-calculator.html'
2. Click 'Calculate'

*Expected: Validation failure***VT-44 | Break mortgage calculator by entering extreme loan term [PASSED]**

Page: /mortgage-calculator.html

Target: cloanterm

Type: Boundary Tests

1. Navigate to '/mortgage-calculator.html'
2. Enter '1000' into 'cloanterm'
3. Click 'Calculate'

Expected: Error message

VT-45 | Rapidly switch between tabs on the financial calculator page [PASSED]

Page: /financial-calculator.html

Target: N/A

Type: Navigation & Cross-Page Tests

1. Navigate to '/financial-calculator.html'
2. Click 'Search'
3. Click 'Search'
4. Click 'Search'

*Expected: No UI errors***VT-46 | Test loan calculator with rapid mode switching and submission [PASSED]**

Page: /loan-calculator.html

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/loan-calculator.html'
2. Click 'Calculate'
3. Click 'Clear'
4. Click 'Calculate'
5. Click 'Search'
6. Click 'Calculate'
7. Click 'Search'

*Expected: No UI errors***VT-47 | Trigger calculation error and verify error clearance [PASSED]**

Page: /loan-calculator.html

Target: c2loantermmonth

Type: Boundary Tests

1. Navigate to '/loan-calculator.html'
2. Enter '0' into 'c2loantermmonth'
3. Click 'Calculate'
4. Enter '12' into 'c2loantermmonth'
5. Click 'Calculate'

*Expected: Error message is cleared***VT-48 | Submit empty form and verify error message [PASSED]**

Page: /mortgage-calculator.html

Target: chouseprice

Type: Validation Enforcement Tests

1. Navigate to '/mortgage-calculator.html'
2. Click 'Calculate'
3. Enter '100000' into 'chouseprice'
4. Click 'Calculate'

*Expected: Error message is displayed and then cleared***VT-49 | Submit with invalid value and verify error clearance [PASSED]**

Page: /interest-calculator.html

Target: cstartingprinciple

Type: Negative Input Tests

1. Navigate to '/interest-calculator.html'
2. Enter '-1' into 'cstartingprinciple'
3. Click 'Calculate'
4. Enter '1000' into 'cstartingprinciple'
5. Click 'Calculate'

Expected: Error message is cleared

VT-50 | Clear all fields after successful calculation and submit emp [PASSED]

Page: /payment-calculator.html

Target: cloanamount

Type: State Persistence Tests

1. Navigate to '/payment-calculator.html'
2. Enter '10000' into 'cloanamount'
3. Enter '12' into 'cloanterm'
4. Enter '5' into 'cinterestrates'
5. Click 'Calculate'
6. Click 'Clear'
7. Click 'Calculate'

*Expected: Error message is displayed***VT-51 | Enter valid data, click Clear/Reset, and verify error message [PASSED]**

Page: /retirement-calculator.html

Target: cagenow

Type: Validation Enforcement Tests

1. Navigate to '/retirement-calculator.html'
2. Enter '30' into 'cagenow'
3. Enter '60' into 'cretireage'
4. Click 'Clear'
5. Click 'Calculate'

*Expected: Error message is displayed***VT-52 | Trigger calculation error and verify error clearance [PASSED]**

Page: /amortization-calculator.html

Target: cloantermmonth

Type: Boundary Tests

1. Navigate to '/amortization-calculator.html'
2. Enter '0' into 'cloantermmonth'
3. Click 'Calculate'
4. Enter '12' into 'cloantermmonth'
5. Click 'Calculate'

*Expected: Error message is cleared***VT-53 | Submit with invalid value and verify error clearance [PASSED]**

Page: /auto-loan-calculator.html

Target: csaleprice

Type: Negative Input Tests

1. Navigate to '/auto-loan-calculator.html'
2. Enter '-1' into 'csaleprice'
3. Click 'Calculate'
4. Enter '20000' into 'csaleprice'
5. Click 'Calculate'

*Expected: Error message is cleared***VT-54 | Clear all fields after successful calculation and submit emp [PASSED]**

Page: /loan-calculator.html

Target: cloanamount

Type: State Persistence Tests

1. Navigate to '/loan-calculator.html'
2. Enter '10000' into 'cloanamount'
3. Enter '12' into 'cloanterm'
4. Enter '5' into 'cinterestrates'
5. Click 'Calculate'
6. Click 'Clear'
7. Click 'Calculate'

Expected: Error message is displayed

VT-55 | Enter valid data, click Clear/Reset, and verify error messag [PASSED]

Page: /interest-calculator.html

Target: cstartingprinciple

Type: Validation Enforcement Tests

1. Navigate to '/interest-calculator.html'
2. Enter '1000' into 'cstartingprinciple'
3. Enter '5' into 'cinterestrate'
4. Click 'Clear'
5. Click 'Calculate'

Expected: Error message is displayed

VT-56 | Trigger calculation error and verify error clearance [PASSED]

Page: /payment-calculator.html

Target: cloanterm

Type: Boundary Tests

1. Navigate to '/payment-calculator.html'
2. Enter '0' into 'cloanterm'
3. Click 'Calculate'
4. Enter '12' into 'cloanterm'
5. Click 'Calculate'

Expected: Error message is cleared

7. Structural Validation Confirmation

All 56 test cases were validated against the UI exploration snapshot. The validation engine verified that:

- ✓ All referenced UI element labels exist in the application snapshot.
- ✓ All referenced page paths exist in the crawled page inventory.
- ✓ No hallucinated or invented element labels were detected.
- ✓ Each test case has a valid structure: ID, goal, steps (min 2), and expected result.

8. Appendix

Snapshot Statistics

Metric	Value
Snapshot File	ai_exploration_snapshot.json
Snapshot Size	65041 bytes
Total Pages in Snapshot	10
Total Input Fields	75
Total Buttons	55
Total Navigation Links	193

Pipeline Configuration

Parameter	Value
AI Model	llama-3.3-70b-versatile (Groq)
Generation Batches	6
Tests per Batch	~10
Batch Delay	30 seconds
Validation Engine	3-Layer Anti-Hallucination
Step Generator	Deterministic (Template-Based)

Output Files Reference

File	Path
Raw Test Cases	data/generated_test_cases.json
Validated Test Cases	data/validated_test_cases.json
Rejected Test Cases	data/rejected_test_cases.json
Gauge Specification	specs/ai_exploration.spec
Step Implementations	step_impl/step_implementation.py
This Report	reports/test_report.pdf